UNIT 3 · TOPIC 3.3

Network Segmentation

Quiz

Instructions: This quiz has 18 multiple-choice questions, 1 point each. Circle the letter of the single best answer for each question.

Name: **Date:** **Class:**

1. Which of the following is a technique used to segment a network?

- ○ A. Installing antivirus software on all endpoint devices
- ○ B. Enabling HTTPS on an organization's public-facing web server
- ○ C. Creating VLANs to separate groups of devices into distinct logical segments
- ○ D. Requiring multi-factor authentication for all user logins

2. A network administrator configures separate VLANs for the HR department and the Engineering department on the same physical switch. What does this configuration primarily accomplish?

- ○ A. It encrypts all traffic traveling between HR and Engineering devices.
- ○ B. It separates the two departments into distinct logical network segments.
- ○ C. It prevents external attackers from reaching either department's devices.
- ○ D. It automatically applies firewall rules to traffic between the two departments.

3. Which of the following best describes a security benefit of network segmentation?

- ○ A. It eliminates the need for encryption because traffic stays within isolated segments.
- ○ B. It improves network performance by reducing the number of nodes in each broadcast domain.
- ○ C. It limits the spread of an attack by containing compromised devices within a single segment.
- ○ D. It prevents insider threats by blocking all employee access to sensitive servers.

4. A hospital network is segmented into three VLANs: one for patient monitoring devices, one for administrative workstations, and one for the billing system. An attacker successfully installs malware on an administrative workstation. Which outcome is most likely as a direct consequence of the segmentation?

- ○ A. The attacker cannot reach patient monitoring devices or the billing system without traversing an inter-VLAN routing control point.
- ○ B. The malware will be automatically detected and removed by the VLAN switch firmware.
- ○ C. The malware will be confined to the workstation's hard drive with no network access at all.
- ○ D. The attacker will gain access to all three VLANs because malware can forge VLAN tags.

5. A company has a flat network where 400 employee workstations, a public web server, and an internal payroll database share the same broadcast domain. The security team's primary concern is a compromised employee workstation reaching the payroll database. Which segmentation approach best addresses this goal?

- ○ A. Move all employees to a single VLAN but increase logging on the payroll database server.
- ○ B. Encrypt all traffic on the flat network using TLS to prevent unauthorized reading of payroll data.

○ C. Separate the web server, employee workstations, and payroll database into distinct VLANs with inter-VLAN traffic controlled by a firewall.

○ D. Apply a uniform strong password policy across all devices so a single compromised credential cannot reach the payroll database.

6. Compared to a flat network, how does VLAN-based segmentation improve network security?

○ A. It enforces 802.1X authentication on all devices, preventing unauthorized users from joining the network.

○ B. It eliminates the need for a firewall because each VLAN functions as its own security perimeter.

○ C. It automatically encrypts intra-VLAN traffic to prevent data interception within a segment.

○ D. It reduces the blast radius of a security incident by limiting which devices can directly communicate with each other.

7. A security architect is comparing two network designs for a corporate campus. Design A uses a single flat network with a hardened perimeter firewall. Design B uses VLAN-based segmentation with per-segment access controls at each inter-VLAN routing point. The architect's primary concern is minimizing the impact of a compromised employee workstation being used by a malicious insider. Which evaluation best supports choosing Design B over Design A?

○ A. Design B is preferred because segmentation eliminates insider threats entirely by preventing all employee-to-server traffic.

○ B. Design A is preferred because a properly configured perimeter firewall is sufficient to contain threats originating from inside the network.

○ C. Design B is preferred because VLAN segmentation forces intra-campus lateral movement through access-controlled routing points, reducing the set of resources reachable from any single compromised device.

○ D. Design A is preferred because VLAN segmentation increases administrative complexity without providing measurable security benefit beyond a perimeter firewall.

8. A network architect proposes layering zero-trust principles on top of existing VLAN boundaries. A colleague argues that VLANs already provide adequate segmentation and zero trust adds unnecessary complexity. Which response most accurately evaluates this disagreement?

○ A. The colleague is correct: VLANs fully enforce zero trust because devices in different VLANs cannot communicate by default.

○ B. The architect is correct because VLANs create coarse-grained perimeters, zero trust adds per-session identity and authorization verification that VLAN boundaries alone cannot enforce.

○ C. The architect is correct because zero trust eliminates the need for VLANs, making network segmentation redundant.

○ D. Both are incorrect: zero trust is an application-layer security model and does not interact with network-layer segmentation.

9. A naval submarine network technician manages three LANs: one for weapons systems, one for official naval operations, and one for crew recreation. The technician must recommend segmentation controls proportional to each LAN's sensitivity level. Which design is most defensible from a security standpoint?

○ A. Merge the weapons and operations LANs into one high-security VLAN and isolate only the recreation LAN, reducing administrative complexity while consolidating the most sensitive systems.

○ B. Connect all three LANs to a central switch without routing restrictions but install an intrusion detection sensor on the weapons LAN only.

○ C. Maintain three fully isolated segments: air-gap or hardware-firewall the weapons LAN from all other traffic, enforce strict access controls between operations and recreation, and treat recreation traffic as untrusted.

○ D. Allow the operations LAN to communicate freely with both the weapons and recreation LANs, since all crew members accessing those LANs hold appropriate clearances.

10. A hotel wants every guest room to reach the internet but no room to reach the device in the room next door, without giving each room its own subnet. Which segmentation technique fits?

○ A. Creating a separate VLAN and routed subnet for every room

○ B. A private VLAN in which each room is an isolated port and only the uplink to the gateway is promiscuous

○ C. An air gap between the guest network and the internet

○ D. Configuring every room's switch port as an 802.1Q trunk

11. Which segmentation technique is enforced on each individual workload by policy, rather than at the switch (Layer 2) or the router (Layer 3)?

○ A. A VLAN

○ B. A routed subnet with an access control list on the router

○ C. A physical air gap

○ D. Zero Trust micro-segmentation, where each workload has its own allow-list enforced by a policy enforcement point

12. A company splits its network into finance, staff and guest VLANs with default-deny rules between them. Which threat does this segmentation do the LEAST to stop?

○ A. Malware spreading from one finance workstation to another finance workstation on the same VLAN

○ B. A guest laptop trying to reach the finance database

○ C. A compromised staff PC scanning the finance subnet

○ D. A guest device trying to open the staff file server

13. A small device plugged into an office wall port sends Dynamic Trunking Protocol frames, and the switch port turns into a trunk that carries every VLAN. Which switch settings would have prevented this, and also the related double-tagging attack?

○ A. Enable spanning-tree portfast on every port and keep VLAN 1 as the native VLAN

○ B. Encrypt every trunk link with TLS

○ C. Set user-facing ports to access mode with switchport nonegotiate, and move each trunk's native VLAN off VLAN 1 to an unused VLAN that carries no hosts

○ D. Put every host on the native VLAN so that no frames need tags

14. In the 2013 Target breach, attackers used an HVAC contractor's stolen credentials to log in to a vendor billing portal, then moved through the internal network to the point-of-sale registers. Which control would most directly have contained the stolen credentials?

○ A. Stronger antivirus on the point-of-sale registers only

○ B. Placing the vendor portal in its own zone with default-deny rules between it and the cardholder data environment

○ C. A larger, faster perimeter firewall at the internet edge

○ D. Encrypting the card database at rest

15. A public web server must be reachable from the internet, but if it is compromised it must not be able to reach internal file shares or the domain controllers. Which design meets both requirements?

○ A. Put the web server on the staff VLAN and rely on its host firewall

○ B. Keep a flat internal network and put the web server behind the same perimeter firewall as everything else

○ C. Air-gap the web server from every other network

○ D. Place the web server in a DMZ separated from the internal network by a further firewall boundary, with no shared credentials or direct trust between DMZ hosts and the internal domain

16. In May 2021, Colonial Pipeline shut down its entire pipeline for about six days, even though investigators found no evidence the ransomware reached the operational technology (OT) that runs the pumps and valves. Which explanation, and which set of layered controls, best fits the lesson of the case?

○ A. The ransomware had encrypted the pipeline's controllers; antivirus on the controllers would have kept it running

○ B. The company had no firewall at all; adding a perimeter firewall would have kept it running

○ C. The company could not verify during the incident that its IT/OT boundary had held; a documented Purdue-style boundary with an OT DMZ (no direct IT-to-OT traffic), continuous monitoring of that boundary, and MFA on remote access would have closed the entry point and let it show OT was isolated

○ D. The company shut down only to speed up ransom negotiations; paying sooner would have avoided the shutdown

17. In a data centre, two web servers share one VLAN and subnet. They should reach the application tier only on TCP 8443 and have no reason to talk to each other. The architect wants layers that fail independently. Which design is best?

○ A. Keep the VLAN and subnet boundary between tiers with default-deny ACLs, add per-workload micro-segmentation allowing only web-to-app on 8443 and no web-to-web traffic, and monitor east-west traffic for unexpected connections

○ B. Collapse all tiers into one VLAN and rely on Zero Trust policy alone, since it makes network boundaries unnecessary

○ C. Replace the inter-tier ACLs with a private VLAN, since isolated ports already enforce port-level rules between tiers

○ D. Rely on antivirus on each server to stop lateral movement

18. A DMZ uses private VLANs, with each server on an isolated port. During a penetration test, a tester on one compromised DMZ server still manages to send traffic to another DMZ server. Which explanation and response are most accurate?

○ A. Private VLANs encrypt traffic between ports, so the tester must have stolen the encryption key; rotate the key

○ B. Isolated ports are meant to talk to each other; move the servers to community ports to stop it

○ C. Private VLANs only work on wireless networks; move the DMZ to Wi-Fi

○ D. Private VLANs are a containment control, not an authentication one, and a host that forges Layer 2 frames or bounces traffic through the gateway can get around them; add port security on the switch, default-deny rules at the gateway, and host-level allow-lists on each server